

Residual Gauge Evolution for Background-Subtracted Z4c

AthenaK TDE Debugging Note

June 7, 2026

1 Purpose

The goal of the background-subtracted Z4c system is to evolve a compact secondary object on top of an analytically known primary black-hole background without allowing truncation error from the primary background to dominate the secondary’s self-gravity. The current debugging question is whether the gauge variables, especially lapse and shift, can be treated in a way that is both internally consistent and stable.

The immediate concern is the following. If the background lapse and shift are prescribed analytically, but the star or secondary black hole requires a different local gauge, does a local residual gauge evolution effectively “change the background” and spoil the subtraction? The answer proposed here is no, provided the background remains a fixed reference state and the gauge equations are written for the residual variables in a background-subtracted form. The residual gauge then changes only the full evolved spacetime, not the analytic background used for subtraction.

2 Residual Variables

Let the Z4c evolved variables be denoted schematically by u . The split is

$$u_{\text{full}}(x, t) = u_{\text{bg}}(x, t) + u_{\text{res}}(x, t). \quad (1)$$

Here u_{bg} is a prescribed analytic reference solution, for example Kerr-Schild Schwarzschild in the black-hole rest frame. It is not evolved numerically. It is recomputed from analytic formulae or otherwise prescribed by the background callback. The residual u_{res} is the only dynamical quantity stored and updated by the residual formulation.

The residual evolution should be

$$\partial_t u_{\text{res}} = \mathcal{R}[u_{\text{full}}] - \mathcal{R}[u_{\text{bg}}], \quad (2)$$

with $u_{\text{full}} = u_{\text{bg}} + u_{\text{res}}$. Equation (2) is an exact change of variables, not a linearization. It retains nonlinear cross terms between the background and the residual because $\mathcal{R}[u_{\text{full}}]$ is evaluated on the full state.

For a large mass ratio this is especially useful: the primary background can be large while the secondary field may be small over most of the domain. Near the secondary, however, the residual may be order unity. The method is still internally consistent there; it is then best understood as a reference metric formulation rather than a perturbative formulation.

3 Gauge Variables

The lapse and shift are split in the same way:

$$\alpha_{\text{full}} = \alpha_{\text{bg}} + \alpha_{\text{res}}, \quad (3)$$

$$\beta_{\text{full}}^i = \beta_{\text{bg}}^i + \beta_{\text{res}}^i. \quad (4)$$

The background gauge variables $\alpha_{\text{bg}}, \beta_{\text{bg}}^i$ are the analytic lapse and shift of the prescribed background. They must not be modified locally by the secondary object.

There are three useful gauge modes:

1. **Pure prescribed background gauge.** Set $\alpha_{\text{res}} = 0$ and $\beta_{\text{res}}^i = 0$, or preserve only selected initial residuals. This makes the full gauge equal to the background gauge except for any explicitly preserved residual components.
2. **Background-subtracted dynamic residual gauge.** Evolve the residual lapse and shift using the difference between the full gauge RHS and the background gauge RHS.
3. **Target-damped residual gauge.** Evolve a background-subtracted dynamic residual gauge, but add damping toward a prescribed residual target, usually for startup only.

4 Background-Subtracted Dynamic Gauge

Let the ordinary full-spacetime gauge equations be written schematically as

$$\partial_t \alpha = \mathcal{G}_\alpha[u, \alpha, \beta^i], \quad (5)$$

$$\partial_t \beta^i = \mathcal{G}_{\beta^i}[u, \alpha, \beta^i]. \quad (6)$$

For example, $\mathcal{G}_\alpha$ may represent a 1+log-like lapse condition and $\mathcal{G}_{\beta^i}$ may represent a Gamma-driver-like shift condition.

The residual gauge evolution should be

$$\partial_t \alpha_{\text{res}} = \mathcal{G}_\alpha[u_{\text{full}}, \alpha_{\text{full}}, \beta_{\text{full}}^i] - \mathcal{G}_\alpha[u_{\text{bg}}, \alpha_{\text{bg}}, \beta_{\text{bg}}^i], \quad (7)$$

$$\partial_t \beta_{\text{res}}^i = \mathcal{G}_{\beta^i}[u_{\text{full}}, \alpha_{\text{full}}, \beta_{\text{full}}^i] - \mathcal{G}_{\beta^i}[u_{\text{bg}}, \alpha_{\text{bg}}, \beta_{\text{bg}}^i]. \quad (8)$$

This is the gauge analogue of Eq. (2). The pure background state is an exact fixed point of the residual system: if $u_{\text{res}} = \alpha_{\text{res}} = \beta_{\text{res}}^i = 0$, then the residual RHS vanishes even if the analytic background is not a stationary solution of the ordinary full gauge condition.

This point is essential. A Kerr-Schild Schwarzschild background is not generally stationary under a generic moving-puncture 1+log/Gamma-driver gauge. If one evolved the full gauge equation directly on top of the background, large gauge motion of the primary black hole would be injected into the residual fields. Equations (7) and (8) subtract that background gauge motion.

5 Why This Does Not Feed Back Into the Background

The background is not a separately evolved physical component. It is a fixed reference field. Local residual gauge evolution changes the full spacetime:

$$(u_{\text{full}}, \alpha_{\text{full}}, \beta_{\text{full}}^i) = (u_{\text{bg}} + u_{\text{res}}, \alpha_{\text{bg}} + \alpha_{\text{res}}, \beta_{\text{bg}}^i + \beta_{\text{res}}^i). \quad (9)$$

It does not change

$$(u_{\text{bg}}, \alpha_{\text{bg}}, \beta_{\text{bg}}^i). \quad (10)$$

Nonlinear feedback from the local gauge choice is still present, but it appears in the residual variables through $\mathcal{R}[u_{\text{full}}] - \mathcal{R}[u_{\text{bg}}]$. That is the correct place for it. What would be inconsistent is evaluating the background subtraction using the full local gauge, for example

$$\mathcal{R}[u_{\text{bg}}, \alpha_{\text{full}}, \beta_{\text{full}}^i], \quad (11)$$

or damping the analytic background variables toward a star- or secondary-BH gauge. Those choices would force the reference background to obey a local gauge prescription that is not its analytic solution.

The consistency rule is therefore:

The background callback always returns the pure analytic background in its own analytic gauge. All local gauge differences belong to the residual state.

6 Target-Damped Residual Gauge

For robustness one may add a target damping term:

$$\partial_t \alpha_{\text{res}} = \mathcal{G}_\alpha[u_{\text{full}}, \alpha_{\text{full}}, \beta_{\text{full}}^i] - \mathcal{G}_\alpha[u_{\text{bg}}, \alpha_{\text{bg}}, \beta_{\text{bg}}^i] - \eta_\alpha(t, x) (\alpha_{\text{res}} - \alpha_{\text{res}}^{\text{tar}}), \quad (12)$$

$$\partial_t \beta_{\text{res}}^i = \mathcal{G}_{\beta^i}[u_{\text{full}}, \alpha_{\text{full}}, \beta_{\text{full}}^i] - \mathcal{G}_{\beta^i}[u_{\text{bg}}, \alpha_{\text{bg}}, \beta_{\text{bg}}^i] - \eta_{\beta^i}(t, x) (\beta_{\text{res}}^i - \beta_{\text{res}}^{i, \text{tar}}). \quad (13)$$

The target fields should be residual targets, not background modifications:

$$\alpha_{\text{res}}^{\text{tar}}(x, t) = w(t, x) (\alpha_{\text{TOV/boost}}(x, t) - \alpha_{\text{bg}}(x, t)), \quad (14)$$

$$\beta_{\text{res}}^{i, \text{tar}}(x, t) = w(t, x) (\beta_{\text{TOV/boost}}^i(x, t) - \beta_{\text{bg}}^i(x, t)). \quad (15)$$

For a star, the target may be the analytic static or boosted TOV gauge used in the initial data. For a secondary black hole, a target should probably be used only as a startup driver, because a permanently prescribed secondary-BH gauge may fight the nonlinear two-body geometry.

A simple time ramp is

$$w(t) = \exp[-(t/\tau)^p], \quad (16)$$

with $\tau \sim 50M$ and $p = 2$ or 4 . A compact spatial window around the secondary can also be used if the target should act only where the secondary gauge is meaningful.

The target-damped scheme remains a gauge prescription. It is not a physical source term, provided it vanishes for the pure background case and is applied only to residual gauge variables.

7 Relation to Current AthenaK Options

The current code already contains the main ingredients:

- `use_analytic_background = true` selects the residual/background path.
- `evolve_gauge_residual = false` prescribes gauge residual RHS to zero. In that mode the full lapse and shift are the analytic background values unless residual components are preserved.

- `preserve_lapse_residual = true` keeps the initial lapse residual while still prescribing its RHS to zero. This is a static residual-gauge target for the lapse.
- `evolve_gauge_residual = true` is intended to compute the full gauge RHS and subtract the background gauge RHS.

The debugging evidence so far suggests that pinning the full lapse to the background value can be too crude for a self-gravitating TOV star. In the short static-Minkowski discriminator, preserving the initial TOV lapse residual substantially reduced early metric drift: $\det \gamma_{\min}$ stayed close to unity and the growth of K -related diagnostics was strongly suppressed.

For a boosted star, the shift residual may also be relevant. The pgen appears to construct a boosted lab-frame lapse and shift for the initial star. If `evolve_gauge_residual = false` and only the lapse residual is preserved, then the boosted-star shift residual is still discarded. A natural extension is therefore a controlled `preserve_shift_residual` or `target_shift_residual` option.

8 Recommended Test Matrix

For the current Schwarzschild radial-infall problem, the recommended order is:

1. **Preserve lapse residual only.** Run with

```
<z4c>
evolve_gauge_residual = false
preserve_lapse_residual = true
diss = 0.5 # or 0.6 as a stability-margin check
damp_kappa1 = 0.0
```

Compare the first non-finite time and the growth of Z4c constraints against the previous Schwarzschild run, which first became non-finite near $t = 80M$.

2. **Pure dynamic residual gauge.** Run with

```
<z4c>
evolve_gauge_residual = true
preserve_lapse_residual = false
```

This tests whether the background-subtracted $1+\log/\Gamma$ -driver path is already stable without target damping.

3. **Target-damped lapse residual.** Add a damping term of the form Eq. (12), with the initial TOV or boosted-TOV lapse residual as the target. First try no spatial window, then a secondary-centered window if needed.
4. **Target-damped shift residual.** If the boosted-star runs remain unstable, add the analogous shift target. This is especially relevant because a boosted star has a nonzero local shift in the lab-frame initial data.
5. **Dissipation and damping margin checks.** Only after the gauge-path discriminator is clear, vary KO dissipation and constraint damping. Dissipation may delay a gauge-driven crash, but it should not be treated as the primary fix if the underlying lapse/shift prescription is inconsistent with the self-gravitating secondary.

9 Diagnostics

Each run should record:

- first non-finite time in `mhd.hst` and `z4c.user.hst`;
- constraint norms C, H, M, Z, Θ versus time;
- $\rho_{\max}$, mass, and star tracker location versus time;
- $\alpha_{\min}$, $\alpha_{\max}$, and $\max |\beta^i|$;
- metric health diagnostics such as $\det \gamma_{\min}$, $\chi_{\min}$, K -related maxima, and bad-metric counts;
- y-reflection and z-reflection symmetry norms on dense slices, with high-density masks.

A useful failure classification is:

If Z4c constraints grow rapidly while density remains finite, and MHD becomes non-finite only after the Z4c diagnostics blow up, the leading failure mode is residual space-time/gauge instability rather than primary fluid dispersal.

10 Applicability to a Secondary Black Hole

The residual formulation can be internally consistent for a secondary black hole as well as for a star. There is no formal small-residual assumption in Eq. (2). However, the practical demands are higher:

- the secondary horizon scale must be resolved by AMR;
- the residual near the secondary can be order unity;
- the gauge must avoid lapse collapse, shift pathologies, and poor horizon treatment;
- the initial data may contain significant constraint violation if it is only a superposition rather than a solved binary-BH data set.

For a secondary black hole, a startup target for lapse and shift can help avoid an abrupt gauge shock, but a permanent analytic target is risky. A better strategy is to ramp the target away and let a background-subtracted residual moving-puncture gauge control the secondary.

11 Implementation Checklist

The safest implementation sequence is:

1. Confirm that background state construction always writes pure analytic background lapse and shift.
2. Confirm that full-state reconstruction uses $\alpha_{\text{full}} = \alpha_{\text{bg}} + \alpha_{\text{res}}$ and $\beta_{\text{full}}^i = \beta_{\text{bg}}^i + \beta_{\text{res}}^i$.
3. Confirm that gauge RHS subtraction evaluates $\mathcal{G}_\alpha[u_{\text{full}}] - \mathcal{G}_\alpha[u_{\text{bg}}]$ and $\mathcal{G}_{\beta^i}[u_{\text{full}}] - \mathcal{G}_{\beta^i}[u_{\text{bg}}]$ with the same derivative operators.

4. Add optional residual target damping only to residual gauge RHS.
5. Keep KO dissipation acting on residual fields, not on the reconstructed full state.
6. Verify pure-background tests: with no star or secondary, residuals should remain at roundoff under the chosen gauge path.
7. Verify static TOV on Minkowski before boosted Schwarzschild infall.

12 Summary

Prescribing the analytic background lapse and shift is consistent only if they remain properties of the fixed reference background. Local gauge changes required by the secondary must be carried by residual lapse and shift variables. A background-subtracted residual gauge evolution provides a mathematically consistent way to let the secondary gauge adjust while keeping the primary background subtraction clean. Target damping of the residual gauge can be used as a startup stabilizer, but it should be introduced as a residual gauge prescription rather than as any modification of the analytic background.